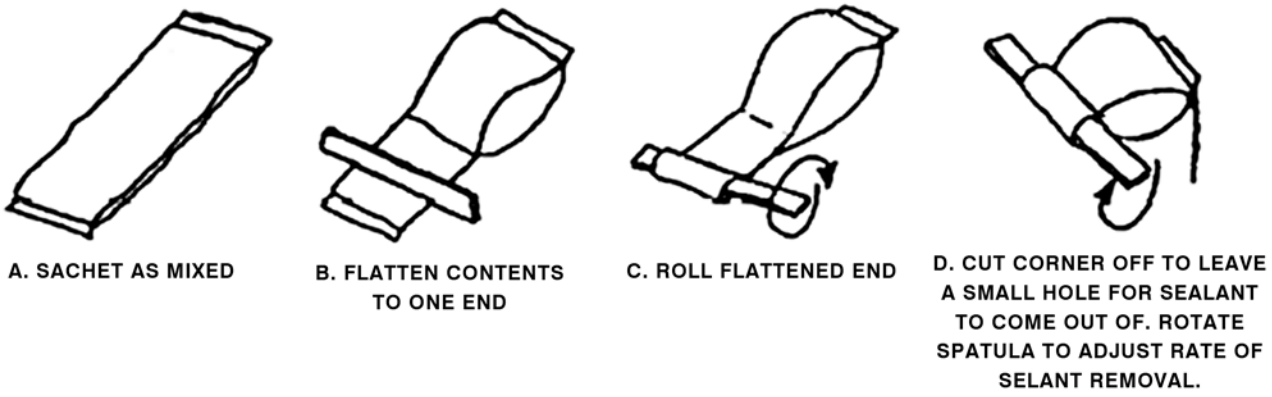
Installation of Aircraft Antennas – General data

1. General

This standard specifies the procedures and requirements for the typical installation of aircraft antenna. To be used when no antenna installation is provided in the applicable sections of the PSM 1-64-2.

2. Materials and Equipment

- Felt pad impregnated with release agent, HR Smith gasket wipes (Product No. 10–900–SS).
- Sealant, MIL–PRF–81733 prepared in accordance to 20-50-01, Weather Sealing – General Data.
- Sealant, Courtaulds Aerospace PR1428 prepared according to 20-50-01, Weather Sealing – General Data. If using sealant which comes in a plastic sachet, after mixing the sealant squeeze it all to one end of the sachet and then cut a small hole in the corner as shown below:
- Sealant spatulas (e.g. SD9164, Mk1, Mk2, Mk3 and Mk4)
- Protective Gloves (e.g. DSC 422–1 and DSC 422–2)
- Approved Safety glasses



Plastic Sachet Sealant
Figure 1

3. Requirements

A. Check that all bonding point surfaces were thoroughly cleaned before the installation of clamps and bonding jumpers. This ensures a satisfactory bond.

B. Check pipe lugs for cracks, bonding terminals for insecure wire braiding, and clamps and connections for looseness.

C. Ensure that all bond connections are tightened to the torque value specified in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data Table 2.

D. Ensure that the method and materials used for bonding and ground connections are in conformance with the standards and procedures in 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data.

E. Ensure that bonding jumpers are installed so that the wire does not come into contact with surrounding structure, equipment, fuel pipes, or other bonding jumpers (refer to 20-75-01, Application of Bonding Jumpers – General data Figure 2). If contact is unavoidable, use a CSP 476 bonding jumper (complete with insulating sleeve). If there is a jumper other than the CSP 476, contact Viking Air Ltd.

F. For cable harnesses with individually shielded wires but without overbraid shields, verify proper shield termination according to PSM 1-64-2W, as applicable, before the backshell is closed.

G. Ensure that touch-up primer and, if applicable, paint top-coat, has been applied to any aircraft structure remaining exposed after installation of all bonding and grounding hardware.

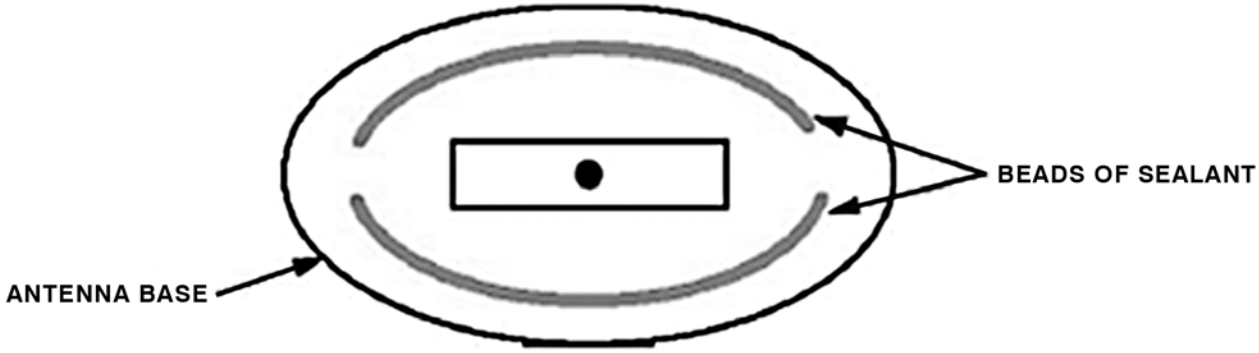
4. Installation Procedures

Note If the antenna is being mounted on the upper surface of the airframe and is fitted with drainage holes, ensure that the drainage holes at the base and sides are kept free of sealant to allow drainage from the shell. Plug and fillet seal any drainage holes on top of the antenna using the plugs provided.

Note If the antenna is being mounted on the lower surface of the airframe and is fitted with drainage holes, ensure that the drainage holes at the tip (i.e., lowest point of inverted antenna) are kept free of sealant to allow drainage from the shell. Plug and fillet seal any drainage holes on the base or sides of the antenna using the plugs provided.

A. For wet-fit and dry-fit gaskets, apply release agent to the antenna base and the mounting area of the airframe using a gasket wipe. Ensure that the surfaces are well covered with release agent, but not to excess.

B. If the engineering drawing specifies use of wet-fit gasket, apply PR1428 sealant to the antenna base and the mounting area of the airframe approximately as shown below and then spread the sealant evenly over the complete surface of the antenna base and the mounting area of the airframe to a thickness of approximately, but not more than, 0.010" using a wooden spatula. Do not apply sealant if installing a dry-fit gasket.

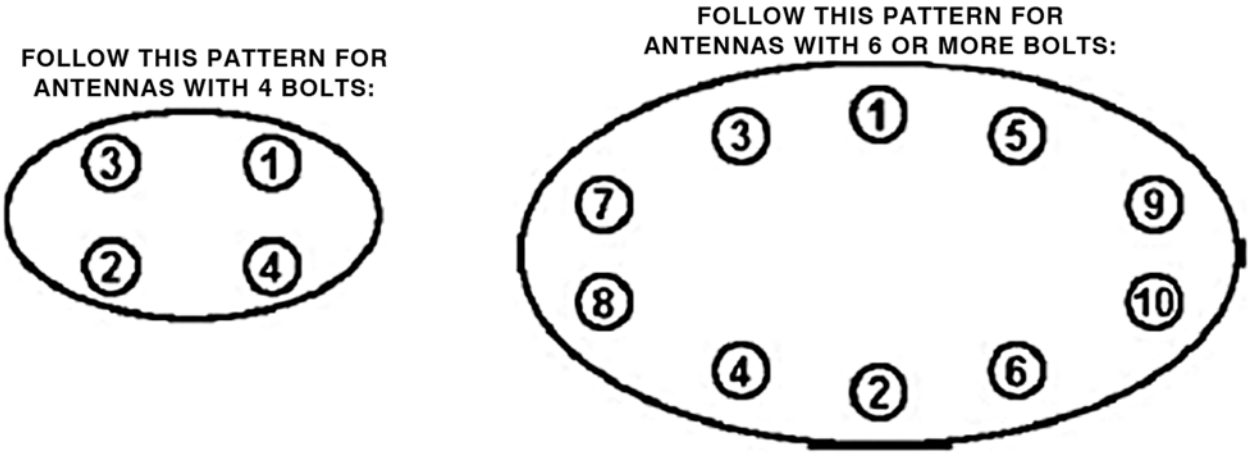


Base of the Antenna
Figure 2

1) Carefully place the gasket on the antenna base, ensuring that the mounting holes in the gasket align with the antenna mounting holes. If a H.R. Smith conductive dry fit sealing gasket is to be used, remove all the release paper from the mating side of the gasket to expose the adhesive surface before applying the gasket.

2) Fit the antenna/gasket assembly on the aircraft and secure it with the fixing bolts. For H.R. Smith conductive dry fit sealing gaskets, remove all the release paper from the mating side of the gasket to expose the adhesive surface before applying in the antenna/gasket assembly.

3) Tighten the bolts, in the pattern shown in Figure 3:



Antenna Bolt Pattern
Figure 3

4) Solvent clean the exposed area of the aircraft structure around the antenna in accordance to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data Para 6.

5) Touch-up the exposed area of airframe around the antenna if required with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer refinishing provides complete coverage of the affected area.

6) If the affected area requires paint touch-up, brush coat the area with the top-coat paint system matching the existing.

7) Use MIL–PRF–81733 sealant to fillet seal around the antenna base in accordance with 20-50-01, Weather Sealing – General Data and seal the securing screws in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data Para 10. The aircraft will be ready to fly approximately 24 hours after sealing.

8) Carry out the antenna electrical bonding check in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data Para 8.

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